

HW # 23

M. Abdullah Siddiqui

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CS-F

Problem 1:

$$-1 = -4$$

$$0 = -6$$

$$1 = 7$$

$$3 = 1$$

$$-2 = -4$$

$$0 = 1$$

$$2 = 2$$

$$1 = 3$$

$$2 = 2$$

$$\text{adj}(A) = \begin{bmatrix} -4 & 1 & 2 \\ -6 & -4 & 3 \\ 7 & 1 & 2 \end{bmatrix}^T = \begin{bmatrix} -4 & -6 & 7 \\ 1 & -4 & 1 \\ 2 & 3 & 2 \end{bmatrix}$$

Problem 2:

$$\begin{aligned} \det(A) &= -1(-4) + 1(7) \\ &= 4 + 7 \\ &= 11 \end{aligned}$$

$$A^{-1} = \frac{1}{11} \begin{bmatrix} -4 & -6 & 7 \\ 1 & -4 & 1 \\ 2 & 3 & 2 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -4/11 & -6/11 & 7/11 \\ 1/11 & -4/11 & 1/11 \\ 2/11 & 3/11 & 2/11 \end{bmatrix}$$

Problem 3:

$$\begin{bmatrix} -1 & 2 & -3 \\ 2 & 0 & 1 \\ 3 & -4 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

$A \quad \quad \quad X = B$

$$\begin{aligned} |A| &= -2(8-12) - 1(4-6) \\ &= 8+2 \\ &= 10 \end{aligned}$$

$$A_1 = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 0 & 1 \\ 2 & -4 & 4 \end{bmatrix} \Rightarrow |A_1| = -1(-4-4) = 8$$

$$\begin{aligned} A_2 &= \begin{bmatrix} -1 & 1 & -3 \\ 2 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} \Rightarrow |A_2| = -2(4+6) - 1(-2-3) \\ &= -20+5 \\ &= -15 \end{aligned}$$

$$\begin{aligned} A_3 &= \begin{bmatrix} -1 & 2 & 1 \\ 2 & 0 & 0 \\ 3 & -4 & 2 \end{bmatrix} \Rightarrow |A_3| = -2(4+4) \\ &= -16 \end{aligned}$$

$$x = \frac{8}{10} = \frac{4}{5}$$

$$y = \frac{-15}{10} = -\frac{3}{2}$$

$$z = \frac{-16}{10} = -\frac{8}{5}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4/5 \\ -3/2 \\ -8/5 \end{bmatrix}$$

Problem 4:

$$\text{Area} = \frac{1}{2} \begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 1 \\ 4 & 3 & 1 \end{vmatrix}$$

$$= \frac{1}{2} [1(-1) - 1(-2) + 1(-2)]$$

$$= \frac{1}{2} (-1 + 2 - 2) = -\frac{1}{2} = \frac{1}{2}$$

Problem 5:

$$\begin{vmatrix} x & y & 1 \\ 2 & 4 & 1 \\ -1 & 3 & 1 \end{vmatrix} = 0$$

$$x(1) - y(3) + 1(10) = 0$$

$$x - 3y + 10 = 0$$

Problem 6:

$$V = \frac{1}{6} \begin{vmatrix} 4 & 0 & 0 & 1 \\ 0 & 4 & 1 & 1 \\ 3 & 5 & 2 & 1 \\ 2 & 2 & 5 & 1 \end{vmatrix}$$

$$= \frac{1}{6} \left[4 \begin{vmatrix} 4 & 1 & 1 \\ 5 & 2 & 1 \\ 2 & 5 & 1 \end{vmatrix} - 1 \begin{vmatrix} 0 & 4 & 1 \\ 3 & 5 & 2 \\ 2 & 2 & 5 \end{vmatrix} \right]$$

$$= \frac{1}{6} \begin{bmatrix} 4(4(-3) - 1(3) + 1(21)) \\ -1(-4(11) + 1(-4)) \end{bmatrix}$$

$$= \frac{1}{6} \begin{bmatrix} 4(-12 - 3 + 21) - 1(-44 - 4) \end{bmatrix}$$

$$= \frac{1}{6} [4(6) + 48]$$

$$= \frac{1}{6} [24 + 48] = \frac{1}{6} \left(\overset{12}{\cancel{72}} \right) = 12$$

Problem 7:

$$\begin{vmatrix} x & y & z & 1 \\ 0 & 1 & 0 & 1 \\ -1 & 3 & 2 & 1 \\ -2 & 0 & 1 & 1 \end{vmatrix} = 0$$

$$x \begin{vmatrix} 1 & 0 & 1 \\ 3 & 2 & 1 \\ 0 & 1 & 1 \end{vmatrix} - y \begin{vmatrix} 0 & 0 & 1 \\ -1 & 2 & 1 \\ -2 & 1 & 1 \end{vmatrix} + z \begin{vmatrix} 0 & 1 & 1 \\ -1 & 3 & 1 \\ 2 & 0 & 1 \end{vmatrix}$$

$$\begin{vmatrix} -1 & 0 & 1 & 0 \\ -1 & 3 & 2 \\ -2 & 0 & 1 \end{vmatrix} = 0$$

$$x \begin{bmatrix} 1(1) + 1(3) \end{bmatrix} - y \begin{bmatrix} 1(3) \end{bmatrix} + z \begin{bmatrix} -1(-3) + 1(-6) \end{bmatrix}$$

$$-1 \begin{bmatrix} -1(3) \end{bmatrix} = 0$$

$$4x - 3y - 3z + 3 = 0$$